
SENTINELX AI

Autonomous Predictive Governance & Regulatory Intelligence Fabric

*"SentinelX transforms compliance from static paperwork
into autonomous operational intelligence."*

Enterprise-grade AI compliance intelligence — built for banks, designed for the future of governance.

THE PROBLEM

Current Banking Compliance Is Broken

Banks continuously receive **RBI circulars**, **SEBI notifications**, **CERT-In advisories**, **NPCI mandates**, **DPDP updates**, and cybersecurity regulations.

Yet the response is entirely **manual**. Legal teams interpret policy without understanding systems. Technical teams understand systems without understanding legal intent.

The result: emails, spreadsheets, meetings, and fragmented ownership.

Core Gap:

Banks lack an intelligent system that can continuously understand, execute, validate, and monitor regulatory compliance.

REACTIVE COMPLIANCE

Issues surface only during audits or after breaches — never before they become critical.

COMPLIANCE DRIFT

Controls silently weaken over time — MFA enabled, then quietly exempted without oversight.

NO OPERATIONAL INTELLIGENCE

Systems track task completion, not real-world governance state. No one sees the full picture.

THE SOLUTION

Introducing SentinelX AI

SentinelX is an **autonomous AI nervous system** for banking governance.

WHAT SENTINELX IS NOT

- ✗ A chatbot
- ✗ A compliance dashboard
- ✗ A regulation summarizer

These are passive tools. SentinelX is fundamentally different.

WHAT SENTINELX IS

An **autonomous governance intelligence platform** that reasons across regulations, departments, systems, and workflows — in real time.

AI AGENTS CONTINUOUSLY

Monitor regulations · **Interpret** legal intent · **Generate** implementation plans
Map organizational impact · **Validate** execution · **Detect** drift · **Forecast** risk

COMPLETE SYSTEM FLOW

End-to-End Workflow



01

REGULATION INGESTION

Continuously scrapes RBI, SEBI, CERT-In, NPCI via web, PDF, API, RSS feeds.

02

SEMANTIC INTERPRETATION

LLMs extract obligations, mandates, and security requirements from legal language.

03

MAP GENERATION

Regulations converted into executable department-level action points.

04

ORGANIZATIONAL MAPPING

Digital Twin identifies affected systems, APIs, owners, and dependencies.

05

COMPLIANCE VALIDATION

AI verifies implementation via Jira, GitHub, IAM policies, SIEM, audit logs.

06

DRIFT DETECTION

Compares expected vs. current governance state continuously.

07

PREDICTIVE GOVERNANCE

Forecasts audit failures, instability, and department-level risk.

08

EXECUTIVE INTELLIGENCE

Leadership receives heatmaps, risk scores, drift alerts, AI summaries.

REGULATORY INTELLIGENCE ENGINE

Continuous Regulation Awareness System

Autonomous, real-time monitoring of India's regulatory ecosystem — so no circular, advisory, or mandate goes unnoticed.

WHAT IT MONITORS

- ▶ RBI circulars
- ▶ SEBI updates
- ▶ CERT-In advisories
- ▶ NPCI notifications
- ▶ AML/KYC guidelines
- ▶ DPDP mandates
- ▶ Cybersecurity frameworks

WHY IT MATTERS

Banks currently depend on manual discovery and email alerts. SentinelX creates **always-on, autonomous regulatory awareness** — nothing slips through.

HOW IT WORKS



PDF INGESTION

Parses regulatory documents automatically



WEB MONITORING

Tracks official portals for new publications



REGULATION CLASSIFICATION

Tags by domain: cybersecurity, fraud, authentication, data



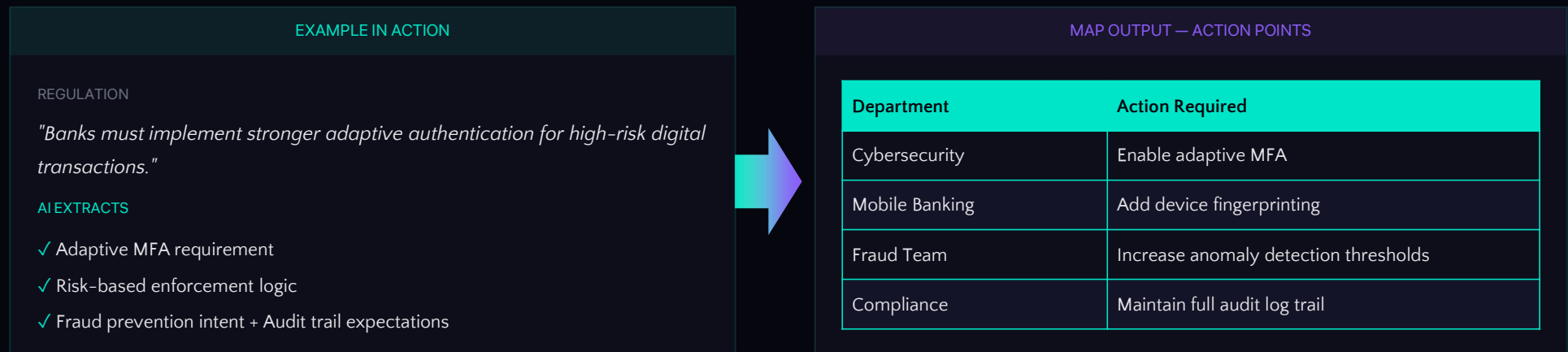
PRIORITY ANALYSIS

Scores by severity, urgency, and affected domains

SEMANTIC INTERPRETATION + MAP ENGINE

Legal Language → Operational Intelligence

The core challenge: Regulatory documents are written in legal language. Translating them into technical action requires human expertise, time, and interpretation — all of which introduce **error and delay**.



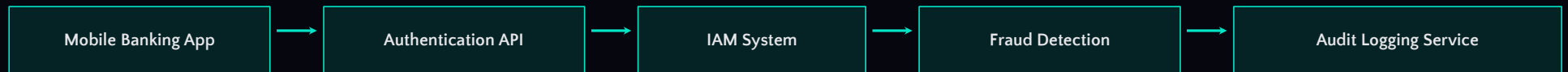
Innovation: SentinelX converts vague policy language into precise, executable operational tasks — automatically.

ORGANIZATIONAL DIGITAL TWIN

AI-Powered Organizational Intelligence

A continuously updated virtual model of the bank — mapping every department, system, API, workflow, IAM dependency, and ownership chain. When a regulation changes, a single policy update can **ripple through multiple systems**.

DEPENDENCY CHAIN EXAMPLE



WHAT THE DIGITAL TWIN ENABLES



DEPENDENCY INTELLIGENCE

Identifies all interconnected systems affected by a regulation



IMPACT ANALYSIS

Predicts downstream operational consequences of changes



OWNERSHIP MAPPING

Automatically surfaces responsible teams for each action



ORGANIZATIONAL REASONING

AI reasons across the full enterprise, not just individual tasks

Key differentiator: Most compliance systems track tasks. SentinelX understands the organization.

COMPLIANCE VALIDATION + DRIFT DETECTION

Continuous Governance Monitoring

COMPLIANCE VALIDATION ENGINE

Verifies that implementation **actually happened** — not just that a ticket was closed.

WHAT IT CHECKS

- ▶ Jira tickets
- ▶ GitHub commits
- ▶ IAM policy states
- ▶ Audit logs
- ▶ SIEM records
- ▶ System configurations

EXAMPLE: MFA VALIDATION

- ✓ MFA policy exists and is active
- ✓ Enforcement enabled across all user groups
- ✓ Authentication logs are being captured

COMPLIANCE DRIFT DETECTION

SENTINELX'S MOST DISTINCTIVE INNOVATION

Drift occurs when compliant systems **gradually degrade** — controls weaken, exceptions accumulate, configurations quietly change.

CLASSIC DRIFT PATTERN

1. MFA enforced at implementation
2. Privileged users exempted "temporarily"
3. Security posture silently weakens
4. **Audit failure** — no one noticed

HOW SENTINELX CATCHES IT

The AI continuously compares **expected governance state** against **current operational reality** — detecting degraded controls before they become violations.

PREDICTIVE GOVERNANCE + REGULATORY MEMORY

AI-Powered Governance Forecasting

PREDICTIVE GOVERNANCE ENGINE

Transforms compliance from **reactive to proactive** — forecasting failures before they occur.

INPUTS THE AI ANALYZES

- ▶ Unresolved MAPs and delayed implementations
- ▶ Accumulated drift signals
- ▶ Department performance velocity
- ▶ Historical violations and audit outcomes
- ▶ Governance trend lines

EXAMPLE OUTPUT

81%

PROBABILITY OF FAILURE (45 DAYS)

Risk factors: Unresolved MFA drift in IAM layer, 3 overdue Fraud Team MAPs.

REGULATORY MEMORY ENGINE

SentinelX learns how your organization **actually behaves** — not just what it reports.

WHAT IT BUILDS OVER TIME

- ▶ Historical regulation and audit response patterns
- ▶ Department-level implementation velocity profiles
- ▶ Recurring governance bottlenecks and weaknesses

EXAMPLE INSIGHT

72%

TREASURY DELAY PROBABILITY

Based on historical compliance velocity across last 8 cycles.

Cumulative advantage: With each regulatory cycle, SentinelX develops deeper organizational behavioral intelligence.

THE AI GOVERNANCE WAR ROOM

DASHBOARD MODULES

Live Regulation Feed

Incoming regulations in real time

MAP Dashboard

Implementation status across departments

Digital Twin Visualization

Interactive enterprise dependency graph

Drift Monitoring Panel

Degraded controls and instability alerts

Predictive Governance Panel

Forecasted audit risk and compliance exposure

Executive Summary Engine

AI-generated governance narratives for leadership

TECHNOLOGY STACK

FRONTEND

Next.js · Tailwind CSS · D3.js · Three.js

BACKEND

FastAPI · Kafka · Redis · PostgreSQL

AI LAYER

Llama 3 · Mistral · LangChain · CrewAI

GRAPH INTELLIGENCE

Neo4j · GraphRAG

BUSINESS IMPACT

♦ Reduce compliance overhead
across all departments

♦ Accelerate implementation
timelines for new regulations

♦ Minimize audit failures
and regulatory penalties

♦ Strengthen cybersecurity
governance posture

♦ Improve operational visibility
enterprise-wide

♦ Shift to proactive governance
from reactive firefighting

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